Introduction to Yule (birth) models

Alex Beams

Department of Mathematics
Simon Fraser University

PhylogeogRaphy Workshop

Day 2

1. Yule model Master equation/Kolmogorov equation
2. Analytical solution using Discrete Laplace Transform (Generating Functions)
3. Tree probabilities

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The Yule (birth) model

We start with the **Yule Model**

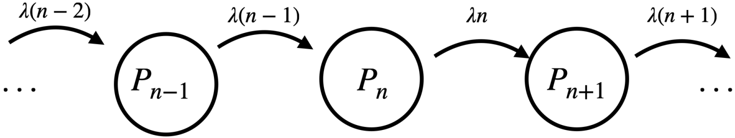
Individuals/species replicate at some rate, but they never die
Specifically, assume the following:

- we start with one individual/species at time $t = 0$
- Each individual/species produces a new individual/species at rate λ
- (So, the time until a lineage gives "birth" is $T \sim \text{Exp}(\lambda)$)

The Yule model

$$\cdots \xrightarrow{\lambda(n-2)} P_{n-1} \xrightarrow{\lambda(n-1)} P_n \xrightarrow{\lambda n} P_{n+1} \xrightarrow{\lambda(n+1)} \cdots$$

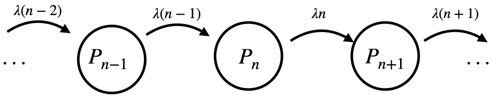
This is a continuous-time Markov chain



Continuous-time Markov chain with a countable state space!

What is the probability the system has n individuals at time t, given that we start with one individual at time $t = 0$?

Master equation:



We need to keep track of the ways to move in and out of the state characterized by n individuals alive at time t

$$\frac{dP_n}{dt} = -n\lambda P_n + (n-1)\lambda P_{n-1}$$

A birth moves us out of P_n to P_{n+1}, and the only way to enter into P_n is from P_{n-1}

Master equation:

$$\frac{dP_n}{dt} = -n\lambda P_n + (n-1)\lambda P_{n-1}$$

The solution to this Master equation is $P_n(t)$, the probability of n individuals at time t. We usually interpret $P_n(t) = \mathcal{P}(X_t = n | X_0 = 1)$, i.e. we start with one individual

Can solve this using (i) generating functions, (ii) guess and check (the solution is a time-dependent geometric distribution), or (iii) mathematical induction[1]

[1] Tanja Stadler et al. *Decoding genomes: from sequences to phylodynamics.* ETH Zurich, 2024.

Where does this equation come from?

$$\frac{dP_n}{dt} = -n\lambda P_n + (n-1)\lambda P_{n-1}$$

This equation encodes information about transition probabilities,

but in disguise. $P_n = \mathcal{P}(X_t = n | X_0 = 1)$, i.e. this is really an equation for the $p_{1,n}$ transition probability:

$$\frac{dp_{1,n}}{dt} = -n\lambda p_{1,n} + (n-1)\lambda p_{1,n-1}.$$

What about the other $p_{i,j}$?

The Chapman-Kolmogorov Equation

Let's derive this rigorously. Our starting point is the Chapman-Kolmogorov (CK) Equation:

$$p_{i,j}(t+s) = \sum_k p_{i,k}(t)p_{k,j}(s)$$

This is **not** a differential equation.

This says: to get from state i to state j over an interval of time $t+s$, we have to account for moves from i to k in the first interval (t), and then moves from k to j in the second interval

For those of you with a background in ODEs and/or dynamical systems: this is a restatement of the semigroup property for flows of a dynamical system in the specific context of Markov chains. (A flow through a point x_0, $\phi_t(x_0)$, has the feature that $\phi_{t+s}(x_0) = \phi_s \circ \phi_t(x_0)$.)

Forward and Backward Equations

$$p_{i,j}(t+s) = \sum_k p_{i,k}(t) p_{k,j}(s),$$

we can **derive** differential equations for the transition probabilities, which allows us to solve for the $p_{i,j}$.

Forward Equation:

$$p_{i,j}(t+\Delta t) = \sum_k p_{i,k}(t) p_{k,j}(\Delta t)$$

Backward Equation:

$$p_{i,j}(t+\Delta t) = \sum_k p_{i,k}(\Delta t) p_{k,j}(t)$$

In the Forward equation, we condition on a jump from k to j near time t. In the Backward equation, we condition on a jump from i to k near time 0.

Forward Equation Derivation for Yule Process

Keep in mind that $p_{i,j}(t) := \mathbb{P}(X_t = j | X_0 = i)$, and that $\lim_{(\Delta t)\to 0} p_{i,j}(\Delta t) = q_{i,j}\Delta t + \mathcal{O}(\Delta t^2)$. We condition on jumps near time t:

$$\begin{aligned} p_{i,j}(t+\Delta t) = &\sum_{k\neq j} p_{i,k}(t) q_{k,j}\Delta t + p_{i,j}(t)(1-q_{j,j}\Delta t) + \mathcal{O}(\Delta t^2), \\ &= p_{i,j-1}(t) q_{j-1,j}\Delta t + p_{i,j}(t)(1-q_{j,j}\Delta t) + \mathcal{O}(\Delta t^2), \\ &= p_{i,j-1}(t)(j-1)\lambda\Delta t + p_{i,j}(t)(1-j\lambda\Delta t) + +\mathcal{O}(\Delta t^2), \\ &\Longrightarrow \end{aligned}$$

$$p_{i,j}(t+\Delta t) - p_{i,j}(t) = -j\lambda p_{i,j}(t) + p_{i,j-1}(t)(j-1)\lambda\Delta t + \mathcal{O}(\Delta t^2).$$

If we divide through by Δt, and then take the limit $\Delta t \to 0$, we obtain $\frac{d}{dt} p_{i,j} = -j\lambda p_{i,j} + (j-1)\lambda p_{i,j-1}$.

Forward Equation Derivation for Yule Process

This Forward Equation

$$\frac{d}{dt}p_{i,j} = -j\lambda p_{i,j} + (j-1)\lambda p_{i,j-1}$$

has a nice property. If we take $X_0 = 1$, so that the Yule Process begins with one individual alive at time 0, we can consider the probability there are n individuals alive at time t, which is described by

$$\frac{d}{dt}p_{1,n} = -n\lambda p_{1,n} + (n-1)\lambda p_{1,n-1}.$$

If we declare that $p_n(t) := p_{1,n}(t)$, we have

$$\frac{dp_n}{dt} = -n\lambda p_n + (n-1)\lambda p_{n-1}.$$

To calculate p_n, we need p_0, p_1, ..., p_{n-1}. If we wrote this as a matrix equation, we would see the system is **triangular**.

Backward Equation Derivation for Yule Process

What happens if we use the Backward derivation instead? Again, keep in mind that $p_{i,j}(t) := \mathbb{P}(X_t = j | X_0 = i)$, and that $\lim_{(\Delta t)\to 0} p_{i,j}(\Delta t) = q_{i,j}\Delta t + \mathcal{O}(\Delta t^2)$. For the Backward equation, the $q_{i,k}$ terms appear first because we condition on jumps near time 0:

$$
\begin{aligned}
p_{i,j}(t+\Delta t) &= \sum_k q_{i,k}\Delta t p_{k,j}(t) + (1 - q_{i,i}\Delta t)p_{i,j}(t) + \mathcal{O}(\Delta t^2), \\
&= q_{i,i+1}\Delta t p_{i+1,j}(t) + (1 - q_{i,i}\Delta t)p_{i,j}(t) + \mathcal{O}(\Delta t^2), \\
&= i\lambda\Delta t p_{i+1,j}(t) + (1 - i\lambda\Delta t)p_{i,j}(t) + \mathcal{O}(\Delta t^2),
\end{aligned}
$$

If we subtract $p_{i,j}(t)$ from both sides, divide through by Δt, and then take the limit $\Delta t \to 0$, we obtain $\frac{d}{dt} p_{i,j} = -i\lambda p_{i,j} + i\lambda p_{i+1,j}$.

Backward Equation Derivation for Yule Process

This Backward equation

$$\frac{d}{dt}p_{i,j} = -i\lambda p_{i,j} + i\lambda p_{i+1,j}$$

does not have the same closure property as the Forward Equation. If we take $X_0 = 1$ as before, and consider $p_{1,n}(t)$ as our variable of choice, we see that

$$\frac{d}{dt}p_{1,n} = -\lambda p_{1,n} + \lambda p_{2,n},$$

and the appearance of $p_{2,n}$ means we need to solve the equation

$$\frac{d}{dt}p_{2,n} = -2\lambda p_{2,n} + 2\lambda p_{3,n},$$

and so on - we have an infinite hierarchy of differential equations.

Backward Equation Derivation for Yule Process

However, we can use the branching process property of the Yule Process to obtain a more compact representation of the Backward Equation. Each lineage in the process produces new lineages independently of the rest. This means that if we start with two lineages ($X_0 = 2$), we can treat these as two independent Yule Processes running in parallel. If one of these lineages produces k individuals by time t and the other produces $n-k$ individuals by time t, the total number of lineages will equal n by time t. We can write this as

$$p_{2,n} = \sum_k p_{1,k} p_{1,n-k}.$$

Setting $p_n(t) = p_{1,n}(t)$ as before, we now have

$$\frac{dp_n}{dt} = -\lambda p_n + \lambda \sum_k p_k p_{n-k}.$$

Yule Process: Forward and Backward Equations

So, our Backward Equation for the Yule Process,

$$\frac{dp_n}{dt} = -\lambda p_n + \lambda \sum_k p_k p_{n-k},$$

is nonlinear. The sum over k is finite, however.

Our Forward Equation,

$$\frac{dp_n}{dt} = -n\lambda p_n + (n-1)\lambda p_{n-1},$$

is linear.

In both cases, we typically define $p_0(0) = 0$, $p_1(0) = 1$.

The **same solution** $p_n(t)$ solves the Forward and Backward equations **simultaneously**. One of the equations is easier to solve.

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Analytic solution using generating functions

We can find analytical solutions to these equations by transforming the Forward Equation using **generating functions**.

The basic idea is to transform the infinite-dimensional system of ODEs in the Forward Equation to a relatively simple partial differential equation (PDE).

Then, we transform the PDE solution back to our original variables to obtain the solution to the Forward Equation.

Transforming the Forward Equation

The Forward equation is:

$$\frac{dp_n}{dt} = -n\lambda p_n + (n-1)\lambda p_{n-1}.$$

To solve, a standard trick is to define a **generating function** $g(z,t) = \sum_{n=1}^{\infty} z^n p_n(t)$.

Think of this as a "bag" to carry all of your p_n terms conveniently.

If you need to look at one of the p_n in particular, do so as follows: $p_n(t) = \frac{\partial^n}{\partial z^n} g(z,t)\big|_{z=0}$.

Transforming the Forward Equation

We derive a partial differential equation (PDE) for $g(z, t)$ from the Forward Equation.

Once we have a formula for $g(z, t)$ strictly in terms of (z, t), we can use our differentiation rule to obtain $p_n(t)$.

First, multiply through by z^n:

$$z^n \frac{dp_n}{dt} = -nz^n \lambda p_n + (n-1) z^n \lambda p_{n-1}.$$

Transforming the Forward Equation

$$z^n \frac{dp_n}{dt} = -nz^n \lambda p_n + (n-1) z^n \lambda p_{n-1}.$$

Next, sum both sides of this equation from $n = 1$ to ∞:

$$\sum_{n=1}^{\infty} z^n \frac{dp_n}{dt} = -\sum_{n=1}^{\infty} nz^n \lambda p_n + \sum_{n=1}^{\infty} (n-1) z^n \lambda p_{n-1}.$$

Now, we use our eyes:

Transforming the Forward Equation

Remember that $g(z,t) = \sum_{n=1}^{\infty} z^n p_n(t)$ and examine these equations:

$$\sum_{n=1}^{\infty} z^n \frac{dp_n}{dt} = -\sum_{n=1}^{\infty} n z^n \lambda p_n + \sum_{n=1}^{\infty} (n-1) z^n \lambda p_{n-1}.$$

Using our definition of $g(z,t)$:

the quantity $\frac{\partial g}{\partial t} = \sum_{n=1}^{\infty} z^n \frac{dp_n}{dt}$

the quantity $\frac{\partial g}{\partial z} = \sum_{n=1}^{\infty} n z^{n-1} p_n(t)$

Transforming the Forward Equation

So this:

$$\sum_{n=1}^{\infty} z^n \frac{dp_n}{dt} = -\sum_{n=1}^{\infty} nz^n \lambda p_n + \sum_{n=1}^{\infty}(n-1)z^n \lambda p_{n-1}.$$

is equivalent to this:

$$\frac{\partial g}{\partial t} = -\lambda z \frac{\partial g}{\partial z} + \lambda z^2 \frac{\partial g}{\partial z}$$

Work through this by hand if you aren't convinced!

Solving the PDE for $g(z,t)$

Next, we have to solve this PDE:

$$\frac{\partial g}{\partial t} = -\lambda z \frac{\partial g}{\partial z} + \lambda z^2 \frac{\partial g}{\partial z}.$$

Let's rearrange this like so:

$$\frac{\partial g}{\partial t} + \lambda z(1-z)\frac{\partial g}{\partial z} = 0.$$

This is a PDE that can be solved using the **Method of Characteristics**.

Method of Characteristics

$$\frac{\partial g}{\partial t} + \lambda z(1-z)\frac{\partial g}{\partial z} = 0.$$

To apply the Method of Characteristics, we see whether a solution of the form $g(z,t) = g(t, z(t))$ could satisfy the PDE. That is, if we apply $\frac{d}{dt}$ to g, the Chain Rule from Calculus would say that

$$\frac{dg}{dt} = \frac{\partial g}{\partial t} + \frac{\partial g}{\partial z}\frac{dz}{dt}.$$

We immediately can see that this will work provided that $\frac{dz}{dt} = \lambda z(1-z)$.

Additionally, the PDE states that $\frac{dg}{dt} = 0$. In other words, $g(z,t) = g_0$, i.e. is constant, along the so-called **characteristic curves** in (z,t) space which satisfy the ODE $\frac{dz}{dt} = \lambda z(1-z)$.

Method of Characteristics

And actually, we can say even more than this. For the Yule Process, we care about the situation where our initial condition is $p_1(0) = 1$, $p_n(0) = 0$ for all $n > 1$. We start with one lineage at time $t = 0$.
Using our definition $g(z, t) = \sum_{n=1}^{\infty} z^n p_n(t)$, plugging in $z = z_0$ and $t = 0$ gives

$$g_0 = g(z_0, 0) = \sum_{n=1}^{\infty} z_0^n p_n(t) = z_0.$$

This means that $g(z, t) = z_0$ along the characteristic curves. We have to find these characteristic curves!! Doing so will give us a formula for z_0 in terms of z and t, i.e. a formula for $g(z, t)$ that we can use to obtain $p_n(t)$ in general.

Solving for the Characteristics

We have an equation for the characteristics. It is

$$\frac{dz}{dt} = \lambda z(1-z).$$

This is a separable ODE. We can do this:

$$\frac{dz}{z(1-z)} = \lambda dt,$$

and then this:

$$\int \frac{dz}{z(1-z)} = \int \lambda dt,$$

Solving for the Characteristics

From this,

$$\int \frac{dz}{z(1-z)} = \int \lambda dt,$$

we do an easy partial fraction decomposition:

$$\int \frac{dz}{z} + \int \frac{dz}{1-z} = \int \lambda dt.$$

Solving for the Characteristics

From here,

$$\int \frac{dz}{z} + \int \frac{dz}{1-z} = \int \lambda dt,$$

we can do this in our sleep:

$$\ln\left(\frac{z}{z_0}\right) - \ln\left(\frac{1-z}{1-z_0}\right) = \lambda t + 0.$$

At time $t = 0$, $z = z_0$, and both sides agree.

Solving for the Characteristics

Rearranging terms, the characteristic curves satisfy the expression

$$\left(\frac{z}{1-z}\right)\left(\frac{1-z_0}{z_0}\right) - e^{\lambda t} = 0.$$

My knee-jerk reaction was to solve for z in terms of t and z_0, the initial condition. But remember: we saw from earlier that $g(z,t) = z_0$ (because $\frac{dg}{dt} = 0$).

So, instead, I solve for z_0:

$$z_0 = g(z,t) = \frac{ze^{-\lambda t}}{1 - z + ze^{-\lambda t}}.$$

This solves the PDE $\frac{\partial g}{\partial t} + \lambda z(1-z)\frac{\partial g}{\partial z} = 0$. Substitute our solution into the PDE to double-check that it satisfies the equation!

Transforming the PDE solution back to $p_n(t)$

$$g(z,t) = \frac{ze^{-\lambda t}}{1 - z + ze^{-\lambda t}}.$$

Now we can find $p_n(t)$, the probability of n lineages at time t. The trick is to use the fact that $g(z,t) = \sum_{n=1}^{\infty} z^n p_n(t)$.

The definition of g ensures that

$$\left.\frac{\partial^n g}{\partial z^n}\right|_{z=0} = n! p_n(t).$$

Transforming the PDE solution back to $p_n(t)$

Here they are for your convenience:

$$\frac{\partial g}{\partial z} = \frac{e^{-\lambda t}}{(1 - z + ze^{-\lambda t})^2},$$

$$\frac{\partial^2 g}{\partial z^2} = \frac{2e^{-\lambda t}(1 - e^{-\lambda t})}{(1 - z + ze^{-\lambda t})^3},$$

$$\frac{\partial^3 g}{\partial z^3} = \frac{6e^{-\lambda t}(1 - e^{-\lambda t})^2}{(1 - z + ze^{-\lambda t})^4},$$

$$\dots$$

$$\frac{\partial^n g}{\partial z^n} = \frac{n!e^{-\lambda t}(1 - e^{-\lambda t})^{n-1}}{(1 - z + ze^{-\lambda t})^{n+1}}.$$

The general formula must be proved by induction (try it!).

Transforming the PDE solution back to $p_n(t)$

The general solution for $p_n(t)$ is

$$p_n(t) = e^{-\lambda t}(1 - e^{-\lambda t})^{n-1}.$$

And obviously: you might as well substitute this into the Forward Equation ODEs dp_n/dt from earlier to verify that this is really the solution.

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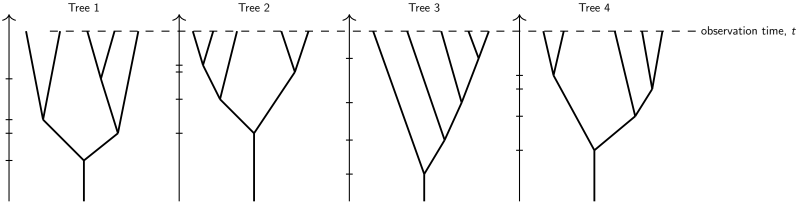
Trees simulated from Yule Process

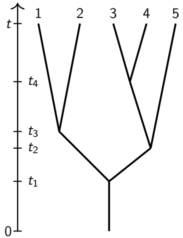
The quantities $p_n(t)$ tell us the probability that the Yule process produces n lineages by time t.

However, there are a lot of ways to get to n lineages by time t:



What are the probabilities of each of these trees, and how are they related to $p_n(t)$?

Towards the probability of a particular tree

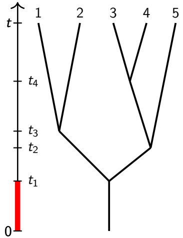


Suppose a tree with n tips by time t has branching times

$$0 < t_1 < t_2 < \cdots < t_{n-1} < t.$$

Let's ignore the tree topology for now - we will come back to it later.

Probability of a set of branching times

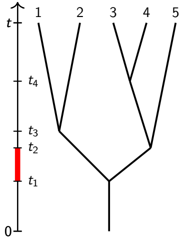


Starting with one lineage:

$$p(t_1) = \lambda e^{-\lambda t_1}.$$

This is just the PDF of branching times for a single lineage in isolation; evaluated at the time t_1, is gives the probability density for observing a branching event at time t_1

Probability of branching times

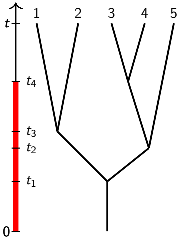


Now, we have two lineages. This means there are two competing exponential rates. The time until the first lineage splits has the PDF given by

$$p(t_2|t_1) = 2\lambda e^{-2\lambda(t_2 - t_1)}.$$

The probability of the tree just after the second branching event (time t_2^+) is the product $p(t_1)p(t_2|t_1) = \left(\lambda e^{-\lambda t_1}\right)\left(2\lambda e^{-2\lambda(t_2-t_1)}\right)$

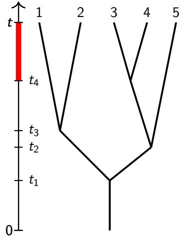
Probability of branching times



The probability of the tree from time $t = 0$ up to the instant after the final split at t_4 is

$$p(t_1|0)p(t_2|t_1)p(t_3|t_2)p(t_4|t_3) =$$
$$\left(\lambda e^{-\lambda t_1}\right)\left(2\lambda e^{-2\lambda(t_2-t_1)}\right)$$
$$\times\left(3\lambda e^{-3\lambda(t_3-t_2)}\right)\left(4\lambda e^{-4\lambda(t_4-t_3)}\right).$$

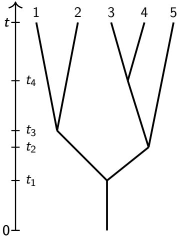
Probability of branching times



But we still have to account for the fact that there was **not** a branching event between time t_4 and time t.

This is given by $e^{-5\lambda(t-t_4)}$.

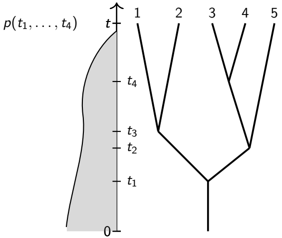
Probability of a particular tree



The PDF of the branching times in general is given by

$$p(t_1, t_2, \dots, t_{n-1}) = \left(\prod_{k=1}^{n-1} k\lambda e^{-k\lambda(t_k - t_{k-1})} \right) e^{-n\lambda(t - t_{n-1})}$$

Probability of a particular tree



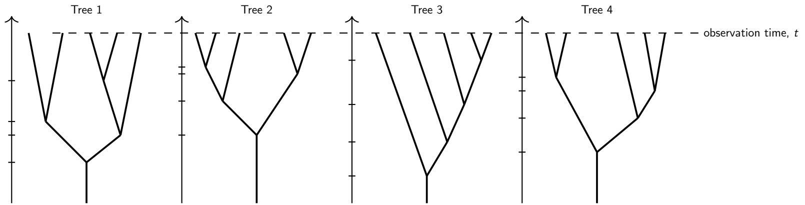
Question: what happens if we integrate over all possible branching times? Do we recover $p_n(t)$?

$$p_n(t) \stackrel{?}{=}$$

$$\int \left(\prod_{k=1}^{n-1} k\lambda e^{-k\lambda(t_k - t_{k-1})} \right) e^{-n\lambda(t-t_{n-1})} dt_1 dt_2 \ldots dt_{n-1}$$

Exercise Show this for p_2 and p_3. If you feel ambitious, prove it for p_n.

Probability of a particular tree



This PDF for the branching times

$$\prod_{k=1}^{n-1} k\lambda e^{-k\lambda(t_k - t_{k-1})}$$

ignores the tree topology. It does not distinguish between different trees with the same branching times.

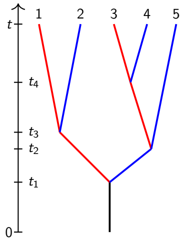
Probability of a particular tree

To account for tree topology, we work recursively on subtrees. It is useful to consider a so-called "oriented" tree, where we distinguish "left" and "right" lineages.

Call the entire tree $\mathcal{T}^0$.

At time t_1, denote the "left" subtree as $\mathcal{T}_l^0$ and the "right" subtree as $\mathcal{T}_r^0$. Then

$$p(\mathcal{T}^0) = \lambda e^{-\lambda t_1} p(\mathcal{T}_l^0) p(\mathcal{T}_r^0)$$



Probability of a particular tree

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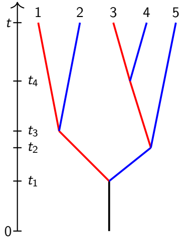


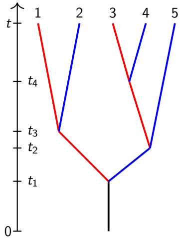
Call the entire tree $\mathcal{T}^0$.

At time t_1, denote the "left" subtree as $\mathcal{T}_l^0$ and the "right" subtree as $\mathcal{T}_r^0$. Then

$$p(\mathcal{T}^0) = \lambda e^{-\lambda t_1} \left(\lambda e^{-\lambda(t_3-t_1)} p(\mathcal{T}_l^{t_3}) p(\mathcal{T}_r^{t_3})\right) \left(\lambda e^{-\lambda(t_2-t_1)} p(\mathcal{T}_l^{t_2}) p(\mathcal{T}_r^{t_2})\right)$$

Probability of a particular tree

Proceeding in this way, we eventually obtain



$$p(\mathcal{T}^0) = \lambda e^{-\lambda t_1} \left(\lambda e^{-\lambda(t_3 - t_1)} e^{-\lambda(t-t_3)} e^{-\lambda(t-t_3)} \right) \left(\lambda e^{-\lambda(t_2-t_1)} \lambda e^{-\lambda(t_4-t_2)} e^{-\lambda(t-t_4)} e^{-\lambda(t-t_4)} e^{-\lambda(t-t_2)} \right),$$
$$= \lambda^4 e^{-\lambda L},$$

where L is the sum of the branch lengths (measured with respect to the t-axis) in the $\mathcal{T}^0$.

Comparing our two formulas

The probability density for a particular tree $\mathcal{T}^0$ with n tips (treating branch times and topologies as given) is

$$p(T^0) = \lambda^{n-1} e^{-\lambda L}.$$

On the other hand, the probability density for a set of branching times (ignoring topology) is

$$p(t_1, \ldots, t_{n-1}) = \prod_{k=1}^{n-1} k\lambda e^{-k\lambda(t_k - t_{k-1})}.$$

However, if you inspect the exponent closely, you may be able to convince yourself that this can be rewritten as

$$p(t_1, \ldots, t_{n-1}) = (n-1)!\lambda^{n-1} e^{-\lambda L}.$$

Discrepancy between our two formulas

This is very interesting. The two formulas differ only by a factor of $(n-1)!$.

$$p(T^0) = \lambda^{n-1} e^{-\lambda L},$$

versus

$$p(t_1, \ldots, t_{n-1}) = (n-1)! \lambda^{n-1} e^{-\lambda L}.$$

The second formula is accounting for the fact that with k lineages at time s s.t. $t_k < s < t_{k+1}$, there are k different ways to get to $k+1$ lineages.

The first formula conditions on the lineages that split at times t_k.

Fitting to data

Another important detail is that the factor $(n-1)!$ does not depend on λ.

If we treat $\mathcal{T}^0$ as given, then we can estimate the parameter λ by maximizing the **likelihood function**

$$\mathcal{L}(\lambda) = p(T^0|\lambda) = \lambda^{n-1}e^{-\lambda L},$$

with respect to λ. Alternatively, we could maximize

$$\mathcal{L}(\lambda) = p(t_1, ..., t_{n-1}) = (n-1)!\lambda^{n-1}e^{-\lambda L}$$

with respect to λ.

The optimal value $\hat{\lambda}$, referred to as the **Maximum Likelihood Estimate (MLE)**, will be the same in either case. (Why?)

Maximum Likelihood Estimate

And just for completeness:

Exercise: use Calculus to find the MLE. Show that it is the same, regardless of which likelihood function you use:

$$\mathcal{L}(\lambda) = p(T^0|\lambda) = \lambda^{n-1}e^{-\lambda L},$$

or

$$\mathcal{L}(\lambda) = p(t_1, ..., t_{n-1}) = (n-1)!\lambda^{n-1}e^{-\lambda L}.$$

Summary

We've learned quite a bit about the Yule Process.

- We derived Forward and Backward Kolmogorov equations for the transition probabilities and related these to $p_n(t)$
- We solved the Forward Equation by using Generating Functions to obtain a formula for $p_n(t)$
- We learned that the PDF of branching times for trees with n lineages at time t marginalizes into $p_n(t)$
- the PDF for a given tree $\mathcal{T}^0$ differs from the PDF of branching times by a factor of $(n-1)!$

Next: we add some death